NOTES (OCTOBER 13)

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Made from handwritten notes by myself with chatGPT and editing.

Review of last month

Let $I = (f_1, \dots, f_n)$ be a finitely generated ideal of a ring A. Write

$$R\Gamma_I A = \Big[A \longrightarrow \prod_i A_{f_i} \longrightarrow \cdots \longrightarrow A_{f_1 \cdots f_n}\Big],$$

where the displayed complex is in cohomological degrees $[0, n]$, in fact has projective amplitude $[0, n]$. $R\Gamma_I(A)$ can also be described as the cofiber of the canonical map $A \to R\Gamma(U, \mathcal{O}_U)$, where U is the complement of $V(I)$. Hence $R\Gamma_I(A)$ and the functors below depends only on $\sqrt{I}$. For $M \in D(A)$, set

$$R\Gamma_I M = R\Gamma_I A \otimes^{\mathbf{L}}_A M, \qquad \widehat{M}^I = R\operatorname{Hom}_A(R\Gamma_I A, M).$$

Both functors $\widehat{(-)}^I$ and $R\Gamma_I(-)$ are idempotent, their essential image denoted $D_{I^\infty\text{-tors}}(A)$ and $D_{I\text{-comp}}(A)$ respectively.

Theorem 1. For $X, Y \in D(A)$,

$$R\operatorname{Hom}_A(\widehat{X}^I, \widehat{Y}^I) = R\operatorname{Hom}_A(R\Gamma_I X, R\Gamma_I Y).$$

Lemma 1.1. For $X, Y \in D(A)$, (the ideal I omitted from the completion)

$$R\operatorname{Hom}_A(X, Y)^\wedge = R\operatorname{Hom}_A(X, \widehat{Y}) = R\operatorname{Hom}_A(\widehat{X}, \widehat{Y}),$$

Consequently, there is an equivalence of categories

$$D_{I\text{-comp}}(A) \underset{(-)^\wedge}{\overset{R\Gamma_I(-)}{\rightleftarrows}} D_{I^\infty\text{-tors}}(A).$$

Theorem 2. For $M \in D(A)$,

$$M \in D_{I\text{-comp}} \iff H^j(M) \in D_{I\text{-comp}}(A) \text{ for every } j.$$

Corollary 2.1. $D_{I\text{-comp}}(A) \cap D^\heartsuit(A)$ is an abelian subcategory.

Nakayama. Let $X \in D_{I\text{-comp}}(A)$, $f \in I$, and

$$C := \operatorname{cofib}(X \xrightarrow{\times f} X).$$

(*i*) If $C = 0$, then $X = 0$.
(*ii*) If $A \in D_{I\text{-comp}}(A)$ and $X \in D^-(A)$, then $C \in D^-_{\text{coh}}(A)$ implies $X \in D^-_{\text{coh}}(A)$.

To prove (*ii*) you resolve X from the right using Theorem 2 and (*i*), cf. [Lyu25, Lemma 5.4].

Matlis duality

In the remainder of today's note, let $(A, \mathfrak{m}, k)$ be a Noetherian complete local ring (in particular, $\mathfrak{m}$ is finitely generated). We omit $\mathfrak{m}$ from the notation when it is obvious (e.g. D_{comp} means $D_{\mathfrak{m}\text{-comp}}$).

We give the axiomatic description of the injective hull of k (cf. [Stacks, Tag 08Y3]).

Theorem 3. There exists an A-module E such that

(*i*) E is injective;
(*ii*) $\mathrm{Hom}_A(k, E) \simeq k$;
(*iii*) E is $\mathfrak{m}^\infty$-torsion (cf. [Stacks, Tag 08XW]).

Note that (*i*) and (*ii*) imply $R\,\mathrm{Hom}_A(k, E) \simeq k$. We will deal with uniqueness later (next time).

Define

$$(-)^\vee := R\,\mathrm{Hom}_A(-, E)$$

so for all $M \in D(A)$ we have a canonical map $\mu_M : M \longrightarrow M^{\vee\vee}$.

Observation. The canonical map $\mu_k : k \to k^{\vee\vee}$ is an isomorphism. Indeed, $\mu_M : M \longrightarrow M^{\vee\vee}$ is nonzero for any $M \in D^\heartsuit(A)$ as soon as $R\,\mathrm{Hom}_A(M, E) \neq 0$, since $\mu_M(x) = (\varphi \mapsto \varphi(x))$ is the evaluation map. But any nonzero map between two isomorphic copies of k is an isomorphism.

Theorem 4. For any $M \in D_{\mathrm{coh}}(A)$, the canonical map

$$\mu_M : M \longrightarrow R\,\mathrm{Hom}_A(R\,\mathrm{Hom}_A(M, E), E)$$

is an isomorphism.

Proof. Since E is injective, we may assume M is bounded. Induct on $d = \dim \mathrm{Supp}\, M$. If $d = 0$, M is a successive extension of shifts of k, so we win by the observation above. Assume $d > 0$ and the theorem is true fo smaller d, take $f \in \mathfrak{m}$ such that

$$\dim(V(f) \cap \mathrm{Supp}\, M) < d.$$

Consider the distinguished triangle which defines C

$$M \xrightarrow{\times f} M \longrightarrow C \longrightarrow +1.$$

Applying double duality gives a morphism of distinguished triangles

$$\begin{array}{ccccccc} M & \xrightarrow{\times f} & M & \longrightarrow & C & \longrightarrow & +1 \\ \downarrow \mu_M & & \downarrow \mu_M & & \downarrow \mu_C & & \\ M^{\vee\vee} & \xrightarrow{\times f} & M^{\vee\vee} & \longrightarrow & C^{\vee\vee} & \longrightarrow & +1. \end{array}$$

We know $\mathrm{Supp}\, C \subseteq V(f) \cap \mathrm{Supp}\, M$. By the induction hypothesis, μ_C is an isomorphism. Thus $\times f$ is an isomorphism on $\mathrm{cofib}(\mu_M)$. However, $M \in D_{\mathrm{comp}}(A)$ as A is complete, and $M^{\vee\vee} \in D_{\mathrm{comp}}(A)$: both $R\,\mathrm{Hom}_A(M, E)$ and E are $\mathfrak{m}^\infty$-torsion, as $M \in D^b_{\mathrm{coh}}(A)$, so Theorem 1 tells us $M^{\vee\vee} \in D_{\mathrm{comp}}(A)$. Hence $\mathrm{cofib}(\mu_M)$ is derived complete. By Nakayama, $\mathrm{cofib}(\mu_M) = 0$. □

Corollary 4.1. $\mathrm{Hom}_A(E, E) = A$.

Remark. There is an equivalence of abelian categories (Matlis duality)

$$\{\text{Noetherian (i.e. finitely generated) } A\text{-modules}\} \underset{(-)^\vee}{\overset{(-)^\vee}{\rightleftarrows}} \{\text{Artinian } A\text{-modules}\}.$$

We have proved part of it ($\mathrm{id} = (-)^{\vee\vee}$ on the category on the left hand side). The rest of the proof is left to the interested reader (not trivial).

Local duality

Definition. The normalized dualizing complex is

$$\omega_A^\bullet := \widehat{E},$$

and we set

$$\mathbb{D}(M) := R\,\mathrm{Hom}_A(M, \omega_A^\bullet) \in D_{\mathrm{comp}}(A).$$

Again uniqueness will be dealt with the next time.

Theorem 5. For $M \in D(A)$,

$$\mathbb{D}(M) \simeq R\operatorname{Hom}_A(R\Gamma_{\mathfrak{m}} M, E) \in D_{\mathrm{comp}}(A).$$

In particular, $\mathbb{D}(k) \simeq k$.

Corollary 5.1. If $M \in D^{[a,b]}(A)$, then $\mathbb{D}(M) \in D^{[-b-\dim A,-a]}(A)$. If $M \in D^{\heartsuit}(A)$, then $\mathbb{D}(M) \in D^{[-\dim A,0]}(A)$. (This is because $\mathfrak{m}$ is the radical of an ideal generated by $\dim A$ elements, hence $R\Gamma_{\mathfrak{m}}(-)$ has cohomological dimension $\dim A$.)

In particular, $\omega_A^{\bullet} \in D^{[-\dim A,0]}(A)$

Proof of Theorem 5.

$$\begin{aligned}
\mathbb{D}(M) &= R\operatorname{Hom}_A(M, \omega_A^{\bullet}) \\
&= R\operatorname{Hom}_A(\widehat{M}, \widehat{E}) && \text{(Lemma 1.1 and Definition)} \\
&= R\operatorname{Hom}_A(R\Gamma_{\mathfrak{m}} M, R\Gamma_{\mathfrak{m}} E) && \text{(Theorem 1)} \\
&= R\operatorname{Hom}_A(R\Gamma_{\mathfrak{m}} M, E) && (E \text{ is } \mathfrak{m}^{\infty}\text{-torsion}).
\end{aligned}$$

□

Theorem 6. For every $M \in D_{\mathrm{coh}}(A)$, the canonical map

$$\gamma_M : M \longrightarrow \mathbb{D}(\mathbb{D}(M))$$

is an isomorphism.

Proof. The proof is the same as for μ_M. Indeed, by Corollary 5.1, we may assume $M \in D^b_{\mathrm{coh}}(A)$. Using $\mathbb{D}(\mathbb{D}(M)) \in D_{\mathrm{comp}}(A)$ (Theorem 5), we can reduce to the case $M = k$. Then Theorem 1 tells us for $X \in D_{\mathrm{comp}} \cap D_{I\text{-tors}}$ we have $\mathbb{D}(X) = X^{\vee}$. Therefore

$$\mathbb{D}(\mathbb{D}(k)) = \mathbb{D}(k^{\vee}) = k^{\vee\vee},$$

and γ_k corresponds to μ_k. □

Theorem 7. If $M \in D_{\mathrm{coh}}(A)$, then $\mathbb{D}(M) \in D_{\mathrm{coh}}(A)$ (in particular, $\omega_A^{\bullet} \in D_{\mathrm{coh}}(A)$).

Proof. By Corollary 5.1, it suffices to consider the case $M \in D^b_{\mathrm{coh}}(A)$. Again, induct on $d = \dim \operatorname{Supp} M$. If $d = 0$, we may assume $M = k$, and $\mathbb{D}(k) \simeq k$. If $d > 0$, take $f \in \mathfrak{m}$ with $\dim(V(f) \cap \operatorname{Supp} M) < d$ and consider the triangle

$$M \xrightarrow{\times f} M \longrightarrow C \longrightarrow +1$$

which defines C. Dualizing gives

$$\mathbb{D}(C) \longrightarrow \mathbb{D}(M) \xrightarrow{\times f} \mathbb{D}(M) \longrightarrow +1.$$

By induction, $\mathbb{D}(C) \in D^b_{\mathrm{coh}}(A)$, and $\mathbb{D}(M) \in D^b_{\mathrm{comp}}(A)$ (Theorem 5 and Corollary 5.1). By Nakayama, $\mathbb{D}(M) \in D_{\mathrm{coh}}(A)$. □

References

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